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Settings

1 - Material Landscape Overview

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Edit dashboard

Goal

Provide a high-level exploration dashboard for the overall material universe. This dashboard acts as the executive and scientific entry point into the platform.

Key Business Questions

- What types of materials exist in the dataset?
- What is the overall distribution of material properties?
- How are crystal systems distributed?
- What percentage of materials are metallic?
- What stability patterns exist

KEY INSIGHTS

Formula vs. Material

- Formula is a "recipe"
- Material is Physical Structure. Even though Diamond and Graphite have the same Formula (just Carbon), they are totally different Materials because their atoms are arranged differently.

Density

We are looking for "Lightweight candidates." A battery material that stores lots of energy but is twice as heavy as another is useless for an electric car.

Band Gap

Energy needed to make a material conduct electricity. It tells us if a material can actually function as part of an electronic or battery circuit. The Rule of Thumb:

- 0 eV: It's a Metal (free-flowing electricity).
- Small Gap (0.1–3 eV): It's a Semiconductor (like silicon, controllable electricity).
- Large Gap (> 4 eV): It's an Insulator (stops electricity, like glass or ceramic).

Metals vs. Non-metallic

"pure conductors" (metals) vs. "functional chemicals" (non-metals) we have. If a specific crystal system is 90% metallic, it might be great for structural steel but terrible for battery storage.

- Metals are great for wires
- Non-metallic oxides (like LFP or NMC) is used for battery cathodes

Crystal Systems

This is the Geometry of the atoms (Cubic, Monoclinic, Hexagonal, etc.). In materials science, Structure = Property:

- Certain shapes (like Cubic) are often more stable and easy to manufacture.
- Certain shapes (like Layered structures) allow ions to move quickly, making them perfect for fast-charging batteries.

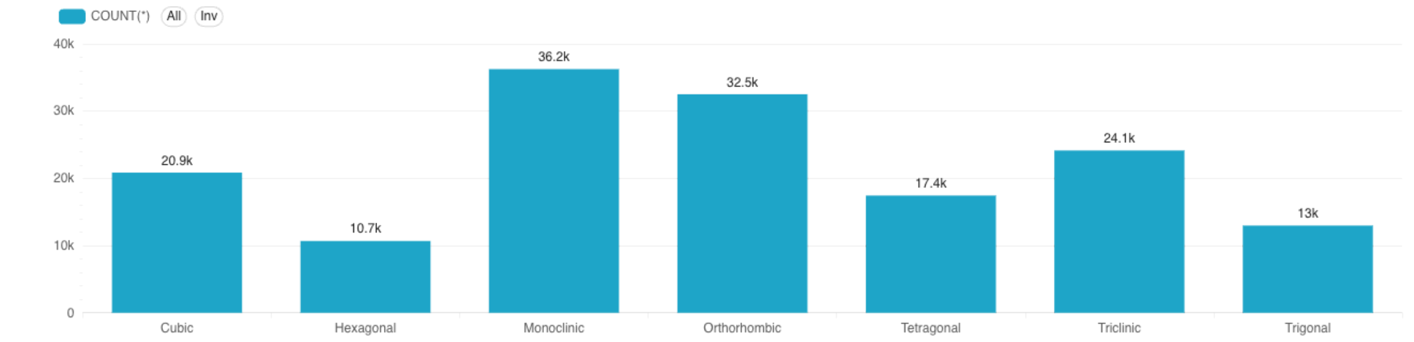
Stability Landscape

Distinguish between materials that could exist in a lab vs. those that are truly stable in the real world. Formation energy tells us if a material is strong, but Energy Above Hull tells us if it is durable and long-lasting:

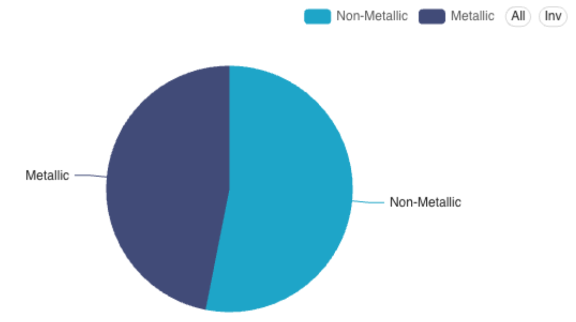
- Energy Above Hull (The "Is there a better option?" metric):
  - 0 eV/atom: The material is on the "Convex Hull." This means it is perfectly stable. It is the absolute "best" way for those elements to arrange themselves.
  - greater than 0 eV/atom: The material is "Metastable." There is a better (lower energy) arrangement out there. Like a ball perched on a ledge, it might eventually "roll down" and decompose into a different material.
- Formation Energy per Atom (The "How much do I want to exist?" metric):
  - The more negative this number is, the more the material "wants" to stay together
- The Bottom Left (Stable & Deep): These are the "Stronghold" materials. They are very hard to break and are thermodynamically perfect. These are your best candidates for mass production.
- The "Hull" Line (Everything at 0 on the Y-axis): These are all the known, stable materials in the universe.
- High Formation Energy but High Hull Energy: These are "Forbidden" materials. They might seem deep, but there's a much deeper hole nearby. They will likely decompose or explode during manufacturing.

Total Formulas	Total Materials	Total Crystal Systems	Avg Band Gap	Avg Density	Metallic Materials %
107k	155k	7	1.06	5.13	46.9%

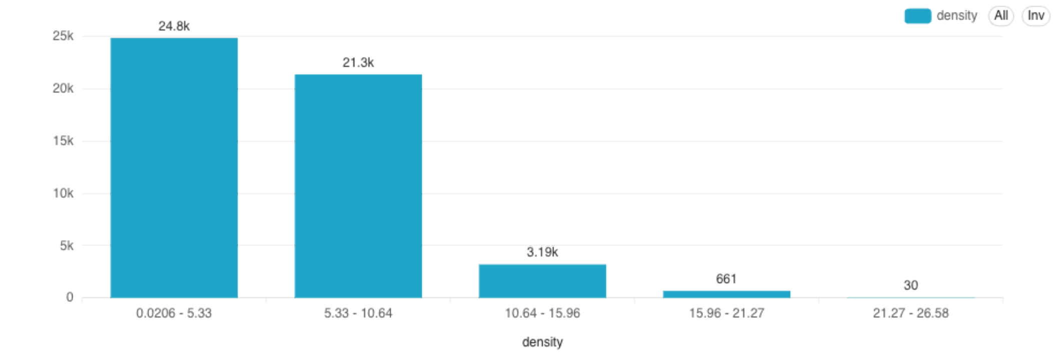
Materials by Crystal System



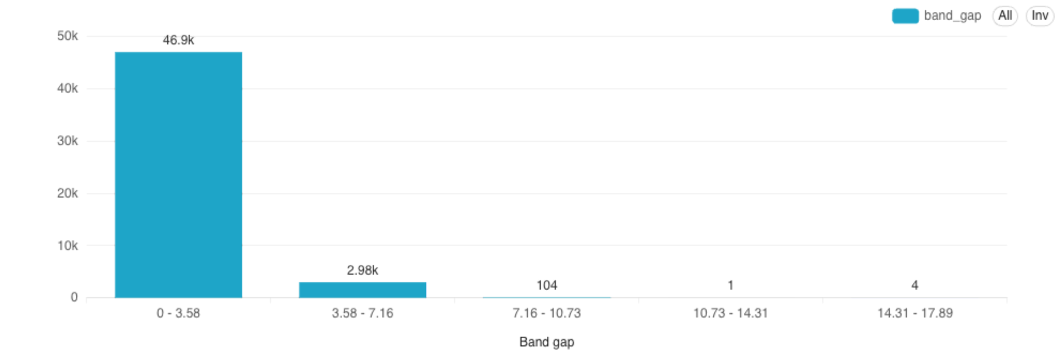
Metallic vs Non-Metallic



Density Distribution



Band Gap Distribution



Stability Landscape